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ORIGINAL ARTICLE

Group-AI Collaboration Enhances Creativity Performance: The Roles of Perspective-Taking and AI Utilisation Strategies

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ABSTRACT

Background: The rise of generative artificial intelligence (AI) in educational research has prompted growing interest in how such tools might support collaborative knowledge construction and creative thinking. While most studies focus on individual-AI interaction, less is known about group-level cognitive and interactional processes when AI is integrated into shared problem-solving settings.

Objectives: This study investigates how group collaboration with AI supports creativity-relevant processes in collaborative learning contexts and explores the psychological mechanisms that mediate this effect.

Methods: Using a within-subject experimental design, 96 university students engaged in a scientific divergent application task under three collaboration modes: group-AI collaboration, individual-AI collaboration and human-only group collaboration. The study adopted a mixed-method approach, combining behavioural coding of perspective-taking and AI utilisation strategies with statistical modelling to examine mediation effects. Tasks were designed to simulate open-ended learning scenarios that emphasise ideation, problem-solving and conceptual integration—core components of collaborative learning.

Results and Conclusions: Group-AI collaboration led to the most creative outputs in terms of usefulness, originality and quality. This effect was mediated by two key mechanisms: strategic use of AI as a generative and exploratory resource and selective, high-quality integration of both peer and AI-generated ideas. While these findings stem from a controlled setting, they offer initial evidence that AI-supported group work may enrich collaborative ideation processes and hold promise for future instructional design and group-based learning practices.

1 | Introduction

Creativity is widely regarded as a core psychological capacity that supports societal progress, technological innovation and complex instructional challenges such as project-based learning and collaborative inquiry in higher education. Although early theories celebrated the 'lone genius', research shows that highly original ideas often arise from group communication and social interaction (Kozlowski and Ilgen 2006). Group creativity, defined as generating novel and useful ideas or solutions, is central to organisational and educational innovation (Lu and Hao 2026; Paulus 2000). Because teamwork is integral to modern work and pedagogy, creative potential is increasingly viewed as emerging from collaborative dynamics. Empirical evidence shows that teams combining diverse perspectives and expertise can outperform the sum of individual contributions (Lu et al. 2021).

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Summary

- What is currently known about this topic?
  - AI can enhance individual creativity in open-ended learning tasks.
  - Most research focuses on individual-AI interaction, not group-AI collaboration.
  - Group creativity depends on cognitive diversity and mutual perspective taking.
  - Effective AI use requires strategic engagement, not passive reliance.
- What does this paper add?
  - Group-AI collaboration outperforms both individual-AI and human-only groups.
  - AI supports perspective taking by enabling selective idea elaboration.
  - Group-AI users adopt more high-quality AI ideas via strategic engagement.
  - AI reshapes team cognition and improves idea evaluation across task stages.
- Implications for practice/or policy.
  - Educators should embed AI into group-based learning, not just individual tasks.
  - Instructional design should promote critical and stage-specific AI use.
  - Training in selective idea adoption may boost team innovation outcomes.
  - Policies should encourage collaborative AI literacy in education systems.

In the 2020s, rapid advances in artificial intelligence (AI) are reshaping human creativity. Generative systems, especially large language models such as ChatGPT, support idea generation and problem solving across domains including education. In classrooms and online courses, they assist writing, brainstorming and interdisciplinary projects. Studies suggest that such systems can reach human level creativity in writing, visual art and music (El Ardeliya et al. 2024; Hitsuwari et al. 2023; Orwig et al. 2024; Zhou and Lee 2024). Researchers, therefore, examine how AI augments human creativity through collaboration (Anantrasirichai and Bull 2022), where AI expands idea diversity and humans select and transform outputs (Sreenivasan and Suresh 2024).

Most research on AI assisted creativity focuses on collaboration between an individual and AI in self-directed learning or workplace tasks. Group-AI collaboration instead positions AI as a generative partner in team problem solving. Studies suggest that generative AI can improve collaborative problem solving and creativity (Wei et al. 2025) and reviews confirm its growing role in AI supported collaborative learning (Kovari 2025). Evidence also shows that interaction structure matters. Creativity is higher when humans alternate with AI rather than only edit AI output (McGuire et al. 2024) and when people generate concepts before requesting AI feedback (Chen and Chan 2024). These findings suggest that treating AI as a partner supports creativity, while heavy reliance on AI ideas can constrain originality.

Individual-AI collaboration also has limits. In product design, Lee and Chung (2024) found no creative gain when participants edited AI ideas rather than using them unchanged, suggesting that benefits depend on user expertise. Medeiros et al. (2025) showed that low-quality AI examples can lower divergent thinking, while highly creative examples do not necessarily outperform human-only teams. In educational settings with limited disciplinary breadth or feedback, learners may therefore struggle to exploit AI effectively. This motivates research on group AI collaboration, where member diversity can support richer prompts, evaluation and idea integration.

Group-AI collaboration involves more than technical interaction and it can engage complex psychological processes. Cognitive-social-motivational models propose that creative performance emerges from cognitive, motivational and social mechanisms (Nijstad and De Dreu 2012; Paulus et al. 2012). These mechanisms are also central in collaborative learning, where co construction of knowledge, shared responsibility and mutual regulation are essential. Introducing AI into group work may alter information processing, motivation such as trust and reliance and communication patterns (Seeber et al. 2020; Zhang et al. 2023). These shifts are highly relevant for technology supported educational collaboration.

This study examines two mechanisms in human-AI collaboration, perspective taking and AI utilisation strategies, that link cognitive, motivational and social aspects of group creativity. Recent work extends perspective taking to human AI settings where people evaluate, adapt or transform AI suggestions (Boers et al. 2025). These processes are also central in collaborative learning, where students integrate multiple viewpoints to co construct knowledge. In human teams, perspective taking reflects willingness to incorporate others' views (Xue et al. 2018). With AI present, it also reflects how users accept or adapt machine generated suggestions. Because AI can surface unconventional ideas by connecting distant concepts, effective collaboration requires openness to novelty and critical judgement to avoid rejecting useful input or accepting weak ideas (Hubert et al. 2024; Rietzschel et al. 2010; Lu and Hao 2026). Accordingly, we treat perspective taking as integration of both peer and AI inputs to examine how AI reshapes idea exchange and creativity.

AI utilisation strategies describe how users incorporate AI tools into the creative process (Heigl 2026). Unlike reciprocal human collaboration, AI use requires deliberate choices about when and how to consult the tool, such as generating ideas, elaborating content or refining and evaluating candidates. These choices matter in professional and educational contexts where learners must flexibly use AI across stages of project-based or inquiry-based learning. Without clear strategies, learners may rely passively on AI outputs, whereas strategic engagement is more likely to treat AI as a partner and improve performance (Boers et al. 2025; Klingbeil et al. 2024). Thus, AI utilisation strategies are central to understanding group AI collaborative processes.

In summary, prior research has established that AI can enhance individual creativity, yet there is limited understanding of how AI shapes collaborative processes at the group level. Studies emphasise the importance of perspective taking and strategic

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engagement in human-AI interactions, but evidence regarding their mediating roles in group contexts remains scarce. Moreover, recent reviews highlight the need to explore how AI can be embedded within collaborative learning environments to optimise idea generation, evaluation and integration. Building on these insights, the present study extends existing work by examining whether group-AI collaboration enhances creative performance compared with individual-AI and human-only collaboration and by clarifying the mediating roles of perspective taking and AI utilisation strategies. Accordingly, we posed the following research questions:

RQ1. *Which of the following collaboration modes demonstrates the strongest positive impact on creative performance: group-AI collaboration, individual-AI collaboration and human-only collaboration?*

RQ2. *How does the level of human perspective taking differ between group-AI and human-only collaboration?*

RQ3. *How does the level of AI-directed perspective taking and the use of AI utilisation strategies differ between group-AI and individual-AI collaboration?*

RQ4. *Do human perspective taking, AI-directed perspective taking and AI utilisation strategies mediate the relationship between collaboration mode and creative performance?*

2 | Methods

2.1 | Participants

The study employed a within-subjects experimental design with one factor: collaborative innovation mode (group-AI vs. individual-AI vs. human-only group). This design was chosen to minimise inter-individual variability and increase statistical power. To address potential order or carry-over effects, the sequence of collaboration conditions and task assignments was fully counterbalanced across participants. A priori power analysis conducted using G*Power 3.1 (effect size $f = 0.25$, $\alpha = 0.05$, power = 0.80) indicated that a minimum of 56 participants was required. A total of 100 participants were recruited via an online questionnaire. Participants were primarily recruited from undergraduate and graduate courses through campus bulletin boards and online learning forums. Inclusion criteria required basic computer literacy but no prior specialised training in AI creativity tools. Exclusion criteria included failure to complete all tasks or non-compliance with instructions. Four participants were excluded for failing to follow task instructions, resulting in a final sample of 96 (64 women; $M_{age} = 21.91$, SD = 3.51). All participants provided informed consent prior to participation. Ethical approval was obtained from the institutional ethics board of the researchers' university (Approval ID: HR2-0252-2024). Each participant received a monetary compensation of 60 RMB upon completing the study.

This study primarily focused on group creativity performance and patterns of interpersonal and human-AI interaction. Accordingly, the dependent variables included (a) creativity performance metrics—usefulness, originality, fluency and high-quality idea fluency; (b) perspective-taking behaviours in both human–human and human–AI interaction and (c) AI utilisation strategies.

2.2 | Procedure

Upon arrival at the laboratory, participants first completed an informed consent form and were informed that their privacy would be strictly protected. All experimental sessions were conducted in a computer laboratory, with participants seated in individual cubicles to minimise distractions. Each session lasted approximately 20 min, including briefing, task performance and debriefing. The experimenter provided standardised written and oral instructions to ensure procedural consistency across conditions. They were then randomly paired to form two-person teams. After the experimenter provided the standardised instructions, participants completed a pre-task questionnaire along with a brief demographic survey. Subsequently, each team was randomly assigned to one of the three collaborative innovation conditions and completed the creativity task by typing their ideas on a computer. In the individual-AI condition, participants collaborated one-on-one with the AI to complete the creativity task. The order of conditions and the assignment of tasks were counterbalanced across participants to mitigate potential order and material confounds.

Collaboration used a split screen setup. Participants communicated via QQ Chat and accessed ChatGPT 4o through a web browser. The two systems were not integrated. QQ Chat served only as the messaging interface, and all AI output came from ChatGPT 4o. Because the model training data and system parameters are proprietary, full transparency is limited and this is a methodological limitation.

At the start of each session, the experimenter posted the creativity task prompt in the QQ chat window. Participants then took turns generating ideas in response to the prompt, contributing one idea per turn during their respective turns (see Figure 1A for the interface layout).

Each participant completed three experimental conditions. In all three conditions, participants generated creative ideas for different scientific divergent application tasks (SDATs), which were of equivalent difficulty and required a high level of prior knowledge. The design of the three collaboration modes (group-AI, individual-AI, human-only group) was informed by prior research in group creativity and human-AI interaction, ensuring conceptual validity. Internal validity was further supported by the within-subjects design, where each participant experienced all conditions, and the order of tasks and conditions was counterbalanced to minimise order and material effects. By equating task difficulty across conditions and holding constant the communication interface, the study controlled for potential confounds and ensured that observed differences could be attributed to collaboration mode rather than extraneous factors. Following classic paradigms in the field of group creativity research, participants in both the group-AI condition and the human-only group condition generated ideas through a turn-taking interaction process. Given that in real-life scenarios, individuals typically have access

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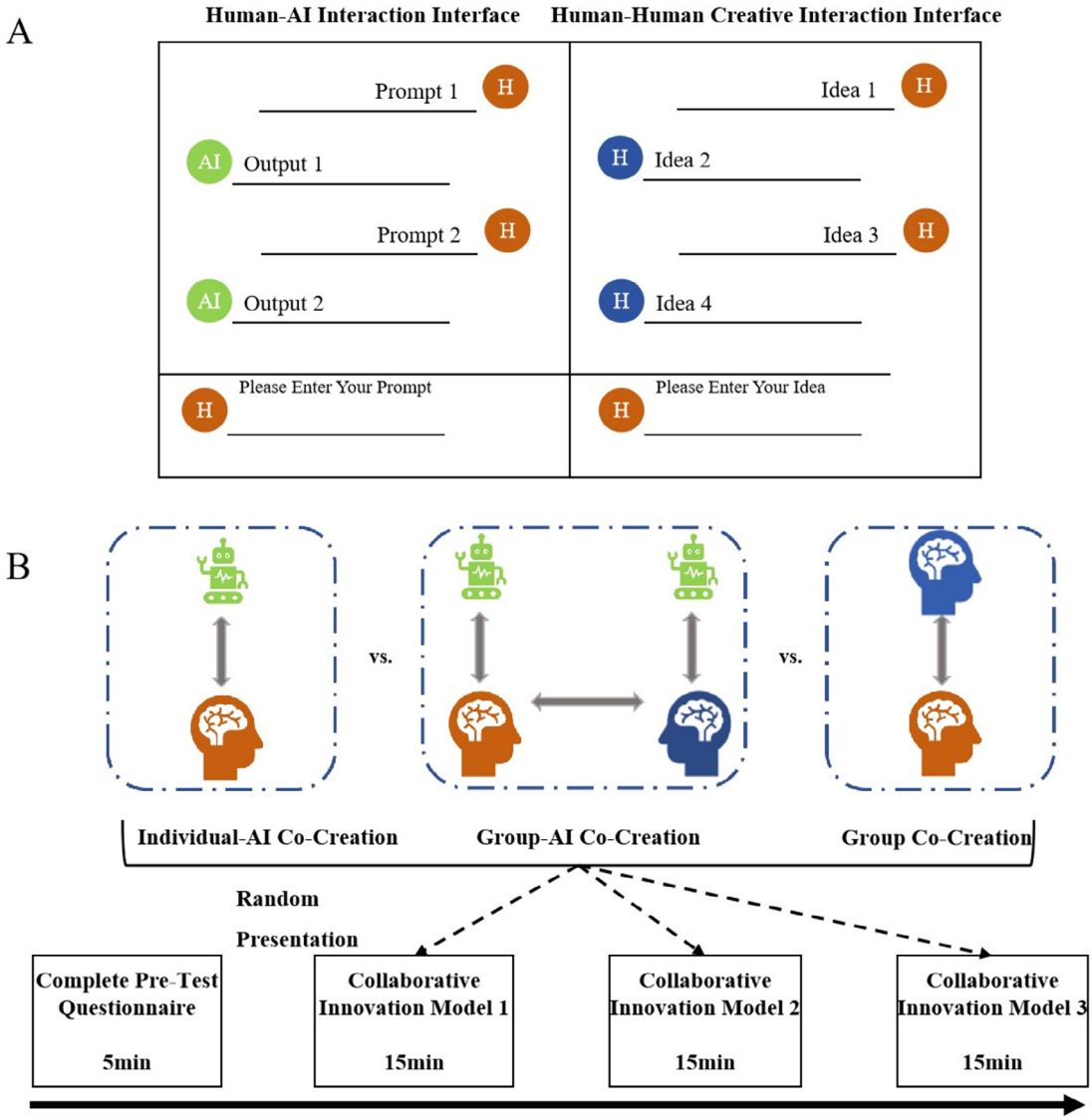


FIGURE 1 | (A) A simplified schematic of the experimental interface. The left panel displays the ChatGPT 4o invocation interface, and the right panel shows the creative communication interface. In the group-AI condition and human-only group condition, the right panel functioned as a public communication platform where participants could view one another's inputs. In the individual-AI condition, the right panel was set as a private communication platform where participants could only see their own input. In the human-only group condition, the AI invocation interface on the left was disabled. (B) Experimental procedure.

to AI tools akin to a personal assistant for handling various tasks, each member in the group-AI collaborative innovation mode was assigned their own AI tool. Moreover, during each participant's own turn to generate ideas, they were allowed to interact with their AI tool, with only one opportunity to consult the AI by asking a single question related to the creativity task. A schematic of the experimental procedure is presented in Figure 1B.

2.3 | Measures

The SDAT was originally developed and validated by Wan et al. (2025) to measure scientific divergent thinking. It consists of 10 items involving various novel materials, such as graphene, carbon fibre and glass microspheres. Each item provides a brief introduction to a specific material and asks participants to generate novel applications for it. The SDAT is particularly effective for our research objectives because it directly assesses divergent ideation in scientific contexts, aligning with the focus of this study on creativity and collaborative knowledge construction. Importantly, we did not modify the original items. Following Wan et al. (2025), we randomly selected three items (transparent ceramics, electronic skin and crystal resin) from the full pool of 10. This random selection ensured equivalence across participants while keeping the experimental workload manageable.

The SDAT has been psychometrically validated in prior large-sample studies. Its internal consistency is high (Cronbach's $\alpha = 0.91$), and confirmatory factor analysis has indicated excellent model fit (CFI = 0.994, TLI = 0.992, RMSEA = 0.030), supporting unidimensionality in measuring scientific creative thinking. Convergent validity has been demonstrated through

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significant correlations with the alternative uses task ($r = 0.52$), scientific creative behaviours and achievements ($r = 0.24$ and 0.20, respectively), openness to experience ($r = 0.19$) and semantic distance indices ($r = 0.24$). Test–retest reliability over a two-month interval is satisfactory ($r = 0.56$), indicating temporal stability. Collectively, these findings confirm the SDAT as a reliable and valid instrument for assessing scientific creative thinking.

The task was evaluated using the following indicators: creative fluency (the number of non-redundant ideas), creative usefulness (the extent to which the ideas could be realistically applied, rated on a 5-point Likert scale; 1 = *not at all useful*, 5 = *very useful*), creative originality (the rarity of the ideas, also rated on a 5-point Likert scale; 1 = *not at all original*, 5 = *very original*) and high-quality creative fluency (the number of ideas rated ≥ 3 on both usefulness and originality). These criteria are consistent with widely used measures of creative performance in divergent thinking research (Guilford 1967; Runco and Jaeger 2012) and with recent applications of the SDAT (Wan et al. 2025). Following Tang et al. (2025) and Wan et al. (2025), who demonstrated the feasibility of using AI models to evaluate creative performance, we assessed creative usefulness and originality in the current study using two AI systems, namely ChatGPT 4o and DeepSeek. The inter-rater reliability (intra-class correlation coefficients, ICCs) between the two AI models was 0.74 for usefulness and 0.83 for originality. To address concerns that generative AI should not replace core research procedures, we additionally conducted human scoring and report it as a transparency benchmark. Three independent human raters (each with more than 2 years of creativity-scoring experience) evaluated all ideas using the same 5-point usefulness and originality rubric; human inter-rater reliability was ICC = 0.74 for usefulness and ICC = 0.71 for originality. We retained AI-based scores as the primary outcome metric because SDAT scoring requires judging scientific plausibility and feasibility (usefulness) as well as substantive novelty (originality) in a knowledge-rich context, which increases rater burden and may lead to greater variability across human judges when the number of ideas is large. Moreover, manual scoring is time-consuming, and prolonged rating work may introduce fatigue effects that shift evaluative criteria over time. This consideration is consistent with the SDAT's methodological motivation to support standardised and scalable evaluation via automated scoring (Wan et al. 2025). It is also aligned with broader evidence showing that large language models substantially improve automated scoring of divergent thinking (Organisciak et al. 2023) and can support automated scoring of both originality and quality in creative problem solving (Luchini et al. 2025). In our data, agreement between human and AI scoring was strong. The correlations between the averaged human ratings and the averaged AI ratings were $r = 0.86$ for usefulness and $r = 0.79$ for originality, indicating that the AI-based scores captured largely the same performance pattern as human ratings while providing a standardised scoring procedure.

Previous research in group creativity has identified individual creative potential and group preference as key predictors of group creative performance (Larey and Paulus 1999; Toh and Miller 2016). Additionally, algorithm aversion and trust in human-AI collaboration have been shown to significantly influence human-AI interaction (Haupt et al. 2025). Therefore, the present study included individual creativity, group preference, algorithm aversion and human–AI trust as control variables in the subsequent analyses.

Individual creative potential was assessed using the Runco Ideational Behaviour Scale (Runco and Jaeger 2012), which consists of 19 items (e.g., 'I have a lot of ideas for new businesses or products'). Responses were rated on a 5-point Likert scale (1 = *strongly disagree*, 5 = *strongly agree*). Higher scores indicate greater ideational potential. Internal consistency was high. The scale demonstrated excellent internal consistency (Cronbach's $\alpha = 0.89$).

Group preference was measured using the Group Preference Scale (Larey and Paulus 1999), which includes 10 items (e.g., 'I enjoy working in groups'). Responses were rated on a 5-point Likert scale (1 = *not at all*, 5 = *very much*), with higher scores indicating a stronger preference for group-based collaboration. The scale demonstrated excellent internal consistency (Cronbach's $\alpha = 0.82$).

Algorithm aversion was assessed using the Algorithm Aversion Scale (Wu et al. 2024), comprising 6 items (e.g., 'I would prefer a human, rather than a mathematical formula, to determine whether I am qualified for a job'). Participants responded using a 5-point Likert scale (1 = *strongly disagree*, 5 = *strongly agree*). Higher scores reflected greater aversion to algorithmic decision-making. The scale demonstrated excellent internal consistency (Cronbach's $\alpha = 0.84$).

Trust in AI was measured using the Human-AI Trust Scale (Gulati et al. 2019), which contains 12 items (e.g., 'I believe AI is competent and effective in creative tasks'). Responses were rated on a 5-point Likert scale (1 = *strongly disagree*, 5 = *strongly agree*), with higher scores indicating greater trust in AI systems. The scale demonstrated excellent internal consistency (Cronbach's $\alpha = 0.91$).

2.4 | Perspective-Taking Coding

Perspective-taking refers to the process by which individuals draw upon ideas from either a teammate or an AI agent during a creative task to generate their own novel ideas. When a participant's idea in the current round belonged to the same conceptual category as their partner's idea from the previous round, this was considered one instance of human-to-human perspective-taking. One point was assigned for each such occurrence. The total human perspective-taking score was calculated by summing all instances of perspective-taking across both participants, with higher scores indicating stronger collaboration between human teammates (Lu et al. 2021).

When a participant's idea in the current round was based on content provided by the AI in the previous round, it was counted as one instance of human-to-AI perspective-taking. One point was assigned for each occurrence. The total AI perspective-taking

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score was defined as the total number of times a participant incorporated AI-generated input, with higher scores reflecting a greater degree of human–AI collaboration.

Experimental observations revealed two additional patterns of perspective-taking. First, in the Group-AI condition, participants sometimes used the AI to elaborate on a teammate's idea or prompted the AI to generate a new idea using the teammate's input. These instances were considered indirect perspective-taking, as they involved integrating both teammate and AI contributions. For such cases, one point was added to both the human and AI perspective-taking scores. Direct human perspective-taking was defined as drawing on a teammate's idea without any AI involvement, whereas indirect human perspective-taking involved using the AI to transform or develop the teammate's idea. The total human perspective-taking score was the sum of direct and indirect instances.

Second, while interacting with the AI, participants either copied the AI's idea verbatim or adapted it to create a new idea. The former was coded as complete AI perspective-taking and the latter as partial AI perspective-taking. The total AI perspective-taking score was the combined total of both types.

Two independent coders categorised all instances of perspective-taking using a coding scheme developed after the experiment. Before formal coding, both raters received training on a subset of transcripts and collaboratively refined the coding manual until full consensus was reached. They then independently coded the entire dataset without knowledge of the study hypotheses. Any discrepancies were discussed and resolved to produce a final agreed-upon coding. Inter-rater reliability was high (Cohen's $\kappa = 0.93$), indicating substantial agreement between raters. The full classification scheme of perspective-taking behaviours is presented in Figure 2A.

The quality of perspective-taking was assessed using the high-quality perspective-taking rate, which was calculated as the rate of adopted ideas that received a score of three or higher on both originality and usefulness. This metric captured the overall quality of idea integration during the task.

2.5 | Coding of AI Utilisation Strategies

After removing two prompts unrelated to the creativity tasks, a total of 209 prompts were collected from participants in the group-AI condition. Based on the Geneplore model (Finke et al. 1992)—which posits that creative thinking involves an initial generative phase of producing novel ideas followed by an exploratory phase of refining and evaluating them—we categorised participants' prompts into two main types. The Geneplore model was chosen because it provides a widely recognised framework for describing creative cognition in two stages, which directly parallels our distinction between prompts that seek novel idea generation (generative strategies) and those that request elaboration, refinement or evaluation (exploratory strategies). This alignment makes the model particularly suitable for operationalising AI utilisation strategies in collaborative creativity.

Generative prompts ($n = 166$) were further classified into three subtypes: broad generative, directional generative and exclusive generative. Exploratory prompts ($n = 43$) were similarly divided into three subtypes: evaluative exploratory, background exploratory and refinement exploratory. The classification of six subtypes was informed by prior research on creativity processes and recent studies on prompting and human–AI interaction (Boers et al. 2025; Heigl 2026). In addition, the subtypes emerged naturally from the linguistic structure of participants' prompts, reflecting how they engaged AI in either a broad, directional, exclusive, evaluative, background-knowledge or refinement-oriented manner.

Generative prompts were further classified into three subtypes according to the way participants phrased their requests. The first subtype, termed broad generative strategy, included prompts that directly asked the AI to generate general ideas, such as 'What are some possible uses for this material?' or 'What novel applications does this material have?' ($n = 84$). The second subtype, labelled directional generative strategy, involved prompts that guided the AI to respond from a particular perspective, such as 'What are some applications of this material in a specific domain?' or 'What properties of this material make it suitable for a particular context?' ($n = 53$). The third subtype was defined as exclusive generative strategy, which encouraged the AI to provide ideas not mentioned earlier, using phrasing such as 'Apart from the ideas already mentioned, what other applications could this material have?' or 'How could this material be used in new and different fields?' ($n = 29$).

Exploratory prompts were also divided into three subtypes. The first, evaluative exploratory strategy, referred to prompts in which participants asked the AI to evaluate the originality or usefulness of an idea, such as 'How novel or useful is this idea?' or 'Can this idea be applied in a specific context?' ($n = 3$). The second, background exploratory strategy, included prompts requesting domain knowledge relevant to the task, such as 'Tell me more about this material' or 'What are the properties of this material?' ($n = 16$). The third subtype, refinement exploratory strategy, referred to prompts that asked the AI to further improve or expand on an existing idea, including questions like 'Based on this idea, can you generate a more novel one?' or 'Can you help develop a more practical version of this idea?' ($n = 24$). For detailed classification, see Figure 2B.

Two independent coders categorised all prompts using the predefined coding scheme. Inter-rater reliability was high (Cohen's $\kappa = 0.92$). Disagreements were resolved through discussion. To assess the validity of this coding scheme, we applied the same classification framework to all prompts collected from the individual-AI condition ($n = 313$). Results indicated that all prompts in the individual-AI condition could be successfully categorised into the six strategies described above, supporting the comprehensiveness of the typology.

In addition, we recorded all timestamps associated with participants' idea generation and AI usage, as well as all textual materials for subsequent analysis.

All statistical analyses were conducted using SPSS Version 26. Mediation analyses were performed using the PROCESS macro.

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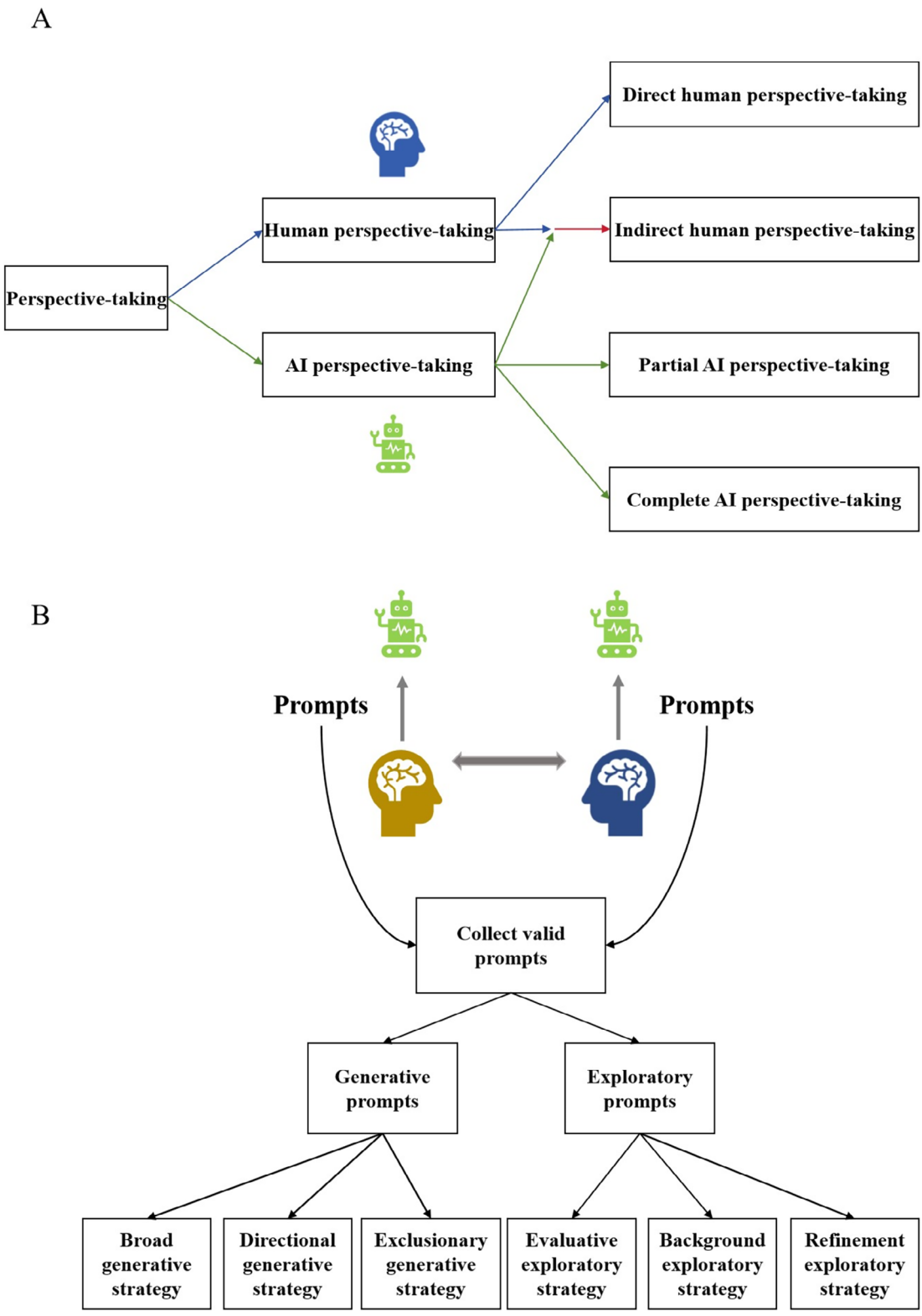


FIGURE 2 | (A) presents a schematic diagram of perspective-taking classifications. (B) Classification framework for AI utilisation strategies.

Mixed-effects models were chosen because they allow simultaneous estimation of fixed effects of collaboration mode and random effects associated with individual differences. Model assumptions (normality, homoscedasticity) were examined before running analyses. For mediation testing, bias-corrected bootstrapping with 5000 samples was employed to provide robust estimates of indirect effects.

3 | Results

3.1 | Effects of Collaboration Mode on Creativity Performance

To examine the effects of different collaboration modes—group-AI, human-only group and individual-AI—on creativity

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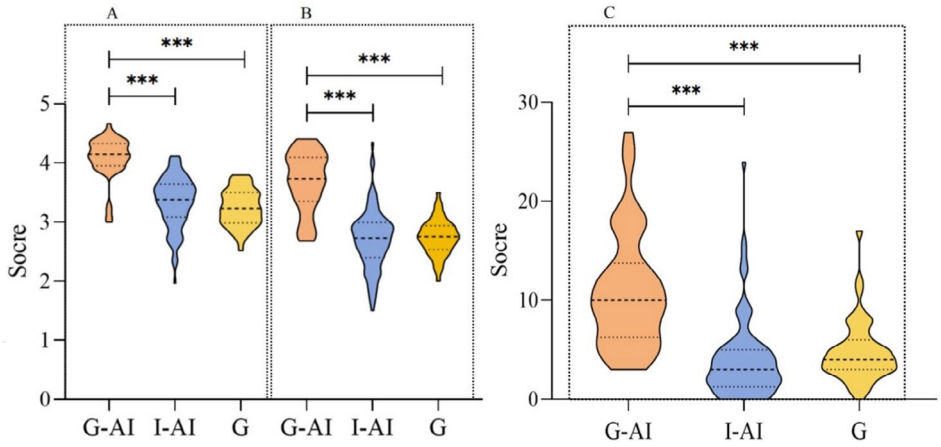


FIGURE 3 | Differences in creative performance across conditions. G-AI, I-AI and G refer to the group-AI, individual-AI and human-only group conditions, respectively. (A–C) Differences in usefulness scores, originality scores and fluency scores of high-quality ideas across conditions, respectively. ***$p < 0.001$.

performance, we conducted mixed-effects model analyses. Subject ID was entered as a random effect to account for individual variability, while collaboration condition was treated as a fixed effect. The primary outcomes included scores on usefulness, originality, fluency and high-quality idea fluency.

Descriptive statistics indicated that participants in the group-AI condition reported higher usefulness scores ($M = 4.09$, SD = 0.35) than those in both the human-only group ($M = 3.24$, SD = 0.31) and individual-AI conditions ($M = 3.34$, SD = 0.43), as shown in Figure 3A. Fixed-effect analyses revealed that usefulness scores in the group-AI condition were significantly higher than in the human-only group condition ($\beta = 0.84$, SE = 0.04, $p < 0.001$) and the individual-AI condition ($\beta = 0.74$, SE = 0.06, $p < 0.001$). No significant difference emerged between the human-only group and individual-AI conditions ($p > 0.05$). The variance for the random effect was 0.09 (SE = 0.09), suggesting that individual-level variability had minimal influence on usefulness scores.

For originality, the group-AI condition again yielded higher scores ($M = 3.68$, SD = 0.50) than the human-only group ($M = 2.74$, SD = 0.30) and individual-AI conditions ($M = 2.70$, SD = 0.50), as illustrated in Figure 3B. Fixed-effect estimates showed that originality scores in the group-AI condition were significantly greater than in the human-only group condition ($\beta = 0.94$, SE = 0.05, $p < 0.001$) and the individual-AI condition ($\beta = 0.98$, SE = 0.07, $p < 0.001$). Again, no significant difference was found between the human-only group and individual-AI conditions ($p > 0.05$). The variance for the random effect was 0.13 (SE = 0.05), indicating low between-subject influence on originality performance.

No significant differences were observed across conditions for fluency scores. However, a significant advantage was found for the group-AI condition in high-quality idea fluency. Participants in this condition generated more high-quality ideas ($M = 11.07$, SD = 5.81) compared to the human-only group ($M = 4.79$, SD = 3.07) and individual-AI conditions ($M = 4.01$, SD = 3.98), as shown in Figure 3C. Fixed-effect analyses confirmed that the group-AI condition outperformed the human-only group condition ($\beta = 6.27$, SE = 0.57, $p < 0.001$) and the individual-AI condition ($\beta = 7.06$, SE = 0.57, $p < 0.001$). No difference was found between the human-only group and individual-AI conditions ($p > 0.05$). The variance for the random effect was 4.40 (SE = 2.10), again suggesting limited influence of individual-level variability.

Taken together, these results indicate that the group-AI collaboration mode significantly enhanced creativity outcomes in terms of usefulness, originality and high-quality idea generation. No significant advantage was observed for either the human-only group or individual-AI conditions across any of the creativity measures. Fluency did not differ significantly between conditions.

3.2 | Effects of Collaboration Mode on Perspective Taking

To examine whether collaboration mode influenced how participants incorporated ideas from others, we first conducted a repeated measures ANOVA comparing human perspective taking between the group-AI and human-only group conditions. The results showed no significant difference in overall human perspective taking between the group-AI condition ($M = 1.40$, SD = 2.27) and the human-only group condition ($M = 1.96$, SD = 1.96), $F(1, 47) = 1.57$, $p = 0.216$. However, a closer look at the type of perspective taking revealed that direct human perspective taking was significantly lower in the group-AI condition ($M = 0.38$, SD = 0.82) than in the human-only group condition ($M = 1.96$, SD = 1.20), $F(1, 47) = 29.51$, $p < 0.001$, $\eta_p^2 = 0.39$ (see Figure 4A).

We further examined the quality of idea adoption by comparing the rate of high-quality human perspective taking between the two conditions. No significant difference was found between the group-AI condition ($M = 0.17$, SD = 0.38) and the human-only group condition ($M = 0.24$, SD = 0.38), $F(1, 47) = 0.91$, $p = 0.345$. These findings suggest that while the overall frequency of collaboration between humans remained similar across conditions, the nature of that collaboration

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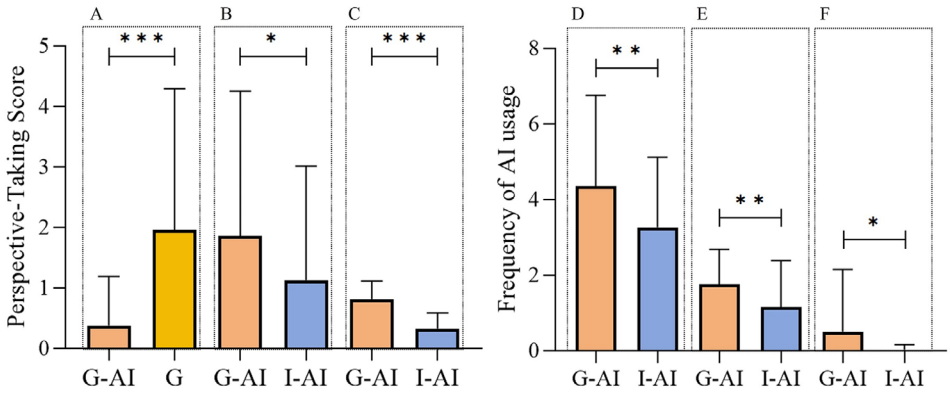


FIGURE 4 | Differences in perspective-taking and AI usage across conditions. (A–C) Differences in perspective-taking: (A) difference in human direct perspective-taking between the G-AI and G conditions; (B) the difference in partial AI perspective-taking between the G-AI and I-AI conditions; (C) the difference in the rate of high-quality AI perspective-taking between the G-AI and I-AI conditions. (D–F) Differences across conditions in AI usage: (D) frequency of AI usage; (E) broad generative strategy and (F) refinement exploratory strategy. *$p < 0.05$; **$p < 0.01$; ***$p < 0.001$.

shifted. In the group-AI condition, participants were more likely to modify their teammate's ideas with the help of AI, rather than adopting them directly. Additionally, across both conditions, participants were generally unlikely to adopt high-quality ideas from their teammates, with the rate hovering around 24%.

We then compared human-AI perspective taking across the group-AI and individual-AI conditions using independent samples *t*-tests. The results revealed no significant difference in overall AI perspective taking between the group-AI condition ($M = 10.23$, SD = 7.15) and the individual-AI condition ($M = 10.50$, SD = 7.52), $t(142) = -0.21$, $p = 0.84$. Similarly, there was no significant difference in complete AI perspective taking between the group-AI condition ($M = 8.38$, SD = 6.91) and the individual-AI condition ($M = 9.38$, SD = 7.62), $t(142) = -0.77$, $p = 0.45$.

However, partial AI perspective taking was significantly higher in the group-AI condition ($M = 1.85$, SD = 2.39) compared to the individual-AI condition ($M = 1.13$, SD = 1.89), $t(142) = 1.99$, $p = 0.048$, $d = 0.33$ (see Figure 4B). When comparing the rate of high-quality ideas among those derived from AI, participants in the group-AI condition ($M = 0.82$, SD = 0.29) significantly outperformed those in the individual-AI condition ($M = 0.33$, SD = 0.26), $t(142) = -10.23$, $p < 0.001$, $d = 1.78$ (see Figure 4C).

Taken together, these findings suggest that while the overall frequency of adopting AI-generated ideas was similar in both collaboration modes, participants in the group-AI condition were more likely to engage in modification and refinement of AI suggestions. Importantly, they were also far more effective in identifying and adopting high-quality ideas, with 82% of their AI-derived ideas meeting the high-quality criteria.

3.3 | Analysis of AI Utilisation Strategies in Human-AI Collaboration

Independent samples *t*-tests were conducted to compare the use of AI strategies between the group-AI and individual-AI conditions. The results showed that participants in the group-AI condition used the AI more frequently ($M = 4.53$, SD = 2.40) than those in the individual-AI condition ($M = 3.26$, SD = 1.85), $t(142) = 3.02$, $p = 0.003$, $d = 0.59$ (see Figure 4D). Use of the broad generative strategy was also significantly higher in the group-AI condition ($M = 1.75$, SD = 0.93) compared to the individual-AI condition ($M = 1.16$, SD = 1.23), $t(142) = 2.95$, $p = 0.004$, $d = 0.54$ (see Figure 4E). In addition, the group-AI condition showed greater use of the refinement exploratory strategy ($M = 0.50$, SD = 1.65) than the individual-AI condition ($M = 0.02$, SD = 0.14), $t(142) = 2.01$, $p = 0.050$, $d = 0.41$ (see Figure 4F).

These results suggest that, compared to individuals working alone with AI, participants in the group-AI condition not only made more frequent use of the AI but also showed a preference for strategies that emphasised broad idea generation and idea refinement. These patterns may reflect more strategic or collaborative engagement with the AI in group contexts.

3.4 | Mediation Analyses: Perspective Taking and AI Utilisation Strategies

To examine whether the effect of group-AI collaboration, compared to the human-only group condition, on creativity performance was mediated by human perspective taking, we conducted a series of mediation analyses using PROCESS Model 4 (Hayes 2013). The independent variable was dummy-coded (0 = human-only group, 1 = group-AI), and the dependent variables were idea usefulness, idea originality and high-quality idea fluency. Direct human perspective taking was included as the mediator.

Results showed that group-AI collaboration had a significant direct effect on idea usefulness ($b = 0.77$, $p < 0.001$, 95% CI [0.63, 0.91]) and a significant indirect effect via human direct perspective taking ($b = 0.08$, 95% CI [0.03, 0.14]), as shown in Figure 5A. Similarly, group-AI collaboration had a significant direct effect on idea originality ($b = 0.86$, $p < 0.001$, 95% CI [0.68, 1.04]) and the indirect effect through human direct perspective taking was also significant ($b = 0.08$, 95% CI [0.01, 0.14]), as shown in Figure 5B. For high-quality idea fluency,

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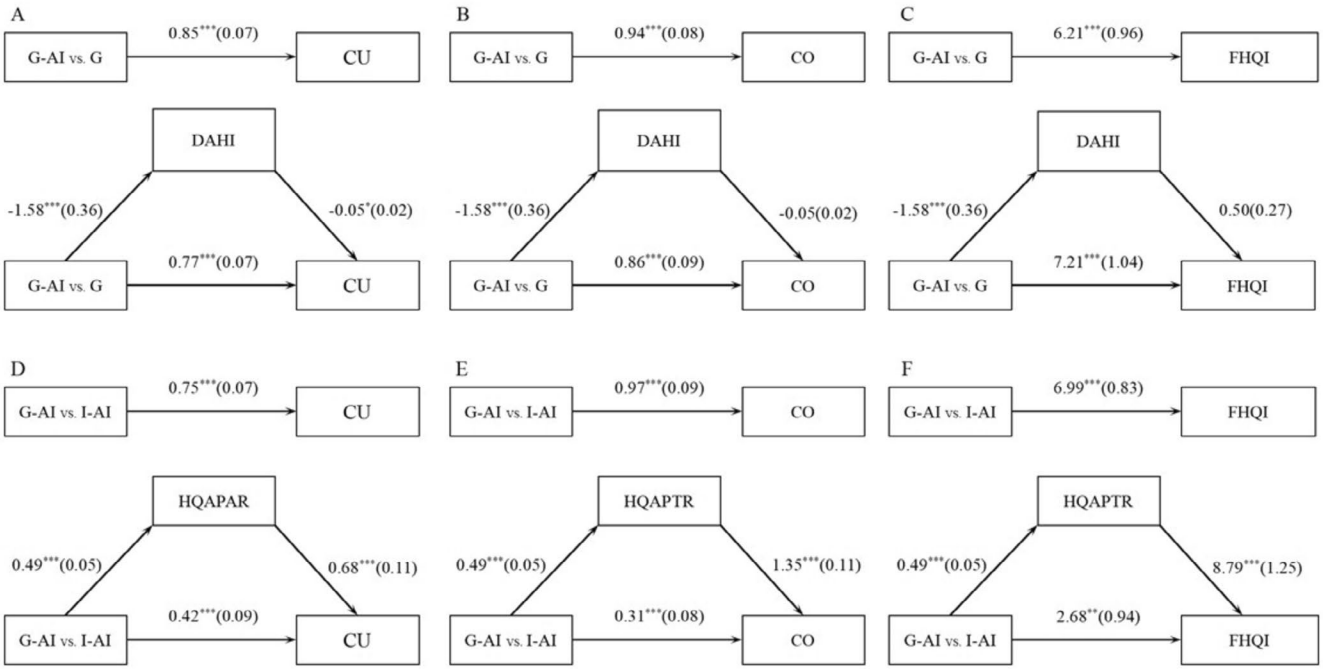


FIGURE 5 | Mediation analysis of perspective taking. Panels A–C show the mediating role of direct human perspective taking in the relationship between collaboration mode (group-AI vs. human-only group) and three creativity outcomes: usefulness (A), originality (B), and high-quality idea fluency (C). Panels D–F show the mediating role of the high-quality AI perspective-taking rate in the relationship between collaboration mode (group-AI vs. individual-AI) and the same three outcomes: usefulness (D), originality (E), and high-quality idea fluency (F). CU, CN and FHQI represent creativity usefulness, creativity originality and fluency of high-quality ideas, respectively. DAHI refers to direct adoption of human ideas and HQAPAR refers to the high-quality AI perspective-taking rate. *$p < 0.05$; **$p < 0.01$; ***$p < 0.001$, with SE in parentheses.

the direct effect of group-AI collaboration was again significant ($b = 7.01$, $p < 0.001$, 95% CI [1.25, 9.07]) and the indirect effect via human direct perspective taking was significantly negative ($b = -0.79$, 95% CI [−1.43, −0.29]), as shown in Figure 5C.

We then tested whether the effect of group-AI collaboration, compared to the individual-AI condition, on creativity performance was mediated by AI perspective taking. The independent variable was dummy-coded (0 = individual-AI, 1 = group-AI) and the mediators included partial AI perspective taking and the rate of high-quality AI perspective taking. Again, idea usefulness, idea originality and high-quality idea fluency served as outcome variables.

The indirect effects through partial AI perspective taking were not significant for any of the outcomes: usefulness ($b = -0.03$, 95% CI [−0.08, 0.01]), originality ($b = -0.02$, 95% CI [−0.06, 0.01]) or high-quality fluency ($b = -0.12$, 95% CI [−0.46, 0.19]). However, the indirect effects through the rate of high-quality AI perspective taking were significant. Group-AI collaboration had a significant direct effect on usefulness ($b = 0.42$, $p < 0.001$, 95% CI [0.25, 0.59]) and a significant indirect effect through high-quality AI perspective taking ($b = 0.33$, 95% CI [0.19, 0.48]), as shown in Figure 5D. Similarly, group-AI collaboration had a significant direct effect on originality ($b = 0.31$, $p < 0.001$, 95% CI [0.15, 0.47]) and a significant indirect effect ($b = 0.66$, 95% CI [0.47, 0.90]), as shown in Figure 5E. For high-quality idea fluency, both the direct effect ($b = 2.68$, $p < 0.001$, 95% CI [0.81, 4.54]) and the indirect effect ($b = 4.31$, 95% CI [2.83, 6.31]) were significant, as shown in Figure 5F.

Finally, we examined whether the effect of group-AI versus individual-AI collaboration on creativity performance was mediated by the use of two specific AI utilisation strategies: broad generative strategy and refinement exploratory strategy. The independent variable was coded as before (0 = individual-AI, 1 = group-AI), and the same three creativity outcomes were used.

Group-AI collaboration had a significant direct effect on usefulness ($b = 0.69$, $p < 0.001$, 95% CI [0.55, 0.84]), and the indirect effect through the broad generative strategy was also significant ($b = 0.06$, 95% CI [0.01, 0.14]), as shown in Figure 6A. The direct effect on originality was also significant ($b = 0.94$, $p < 0.001$, 95% CI [0.76, 1.12]), and the indirect effect via broad generative strategy was significant as well ($b = 0.04$, 95% CI [0.01, 0.13]), as shown in Figure 6B. For high-quality idea fluency, the direct effect was significant ($b = 6.34$, $p < 0.001$, 95% CI [4.71, 7.97]), and the indirect effect via broad generative strategy remained significant ($b = 0.65$, 95% CI [0.17, 1.95]), as shown in Figure 6C.

Regarding the refinement exploratory strategy, group-AI collaboration had a significant direct effect on usefulness ($b = 0.73$, $p < 0.001$, 95% CI [0.58, 0.88]), and a small but significant indirect effect via this strategy ($b = 0.02$, 95% CI [0.01, 0.05]), as shown in Figure 6D.

In summary, compared to the human-only group condition, group-AI collaboration reduced the level of direct human perspective taking, which in turn contributed to enhanced creativity performance in terms of usefulness, originality and high-quality fluency. Compared to the individual-AI condition, group-AI participants demonstrated a stronger tendency to

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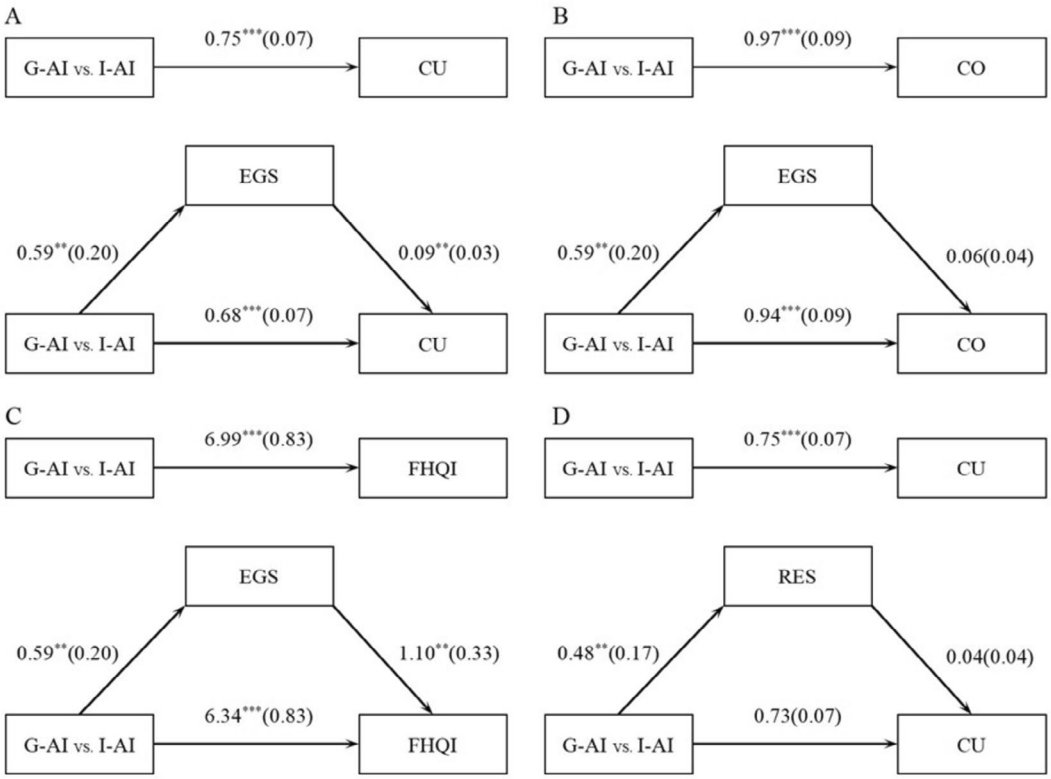


FIGURE 6 | Mediation analysis of AI utilisation strategies. Panels A–C show the mediating role of the broad generative strategy in the relationship between collaboration mode (group-AI vs. individual-AI) and three creativity outcomes: usefulness (A), originality (B), and high-quality idea fluency (C). Panel D shows the mediating role of the refinement exploratory strategy in the relationship between collaboration mode (group-AI vs. individual-AI) and creativity usefulness. CU, CN and FHQI represent creativity usefulness, creativity originality and fluency of high-quality ideas, respectively. EGS refers to the broad generative strategy and RES refers to the refinement exploratory strategy. $^{*}p < 0.05$; $^{**}p < 0.01$; $^{***}p < 0.001$, with SE in parentheses.

selectively adopt high-quality AI-generated ideas, which further improved their creative outcomes. Moreover, participants in the group-AI condition were more likely to engage with AI through broad generative strategies and refinement-oriented strategies, both of which played mediating roles in enhancing creativity performance.

3.5 | Supplementary Analyses

An independent samples *t*-test revealed that participants in the group-AI condition interacted with the AI significantly more frequently ($M = 4.35$, SD = 2.40) than those in the individual-AI condition ($M = 3.26$, SD = 1.85), $t(142) = 3.02$, $p = 0.01$, $d = 0.51$. As previously reported, there was no significant difference between the two conditions in terms of the overall level of AI perspective taking. This pattern suggests that participants in the group-AI condition were more selective and perhaps more critical of the AI's ideas, prompting more frequent queries to elicit better input.

This interpretation is supported by the observation that the group-AI condition also exhibited a notably high rate of 82% for high-quality AI perspective-taking. These findings suggest a pattern of strategic filtering, in which participants actively screened AI outputs to extract only the most useful and novel ideas.

We also conducted a supplementary analysis comparing the time participants spent between initiating an AI query and generating a creative idea (measured in seconds). Results indicated that participants in the group-AI condition responded more quickly ($M = 55.91$, SD = 21.72) than those in the individual-AI condition ($M = 68.48$, SD = 50.85), $t(142) = -2.03$, $p = 0.044$, $d = -0.32$.

Taken together, these results indicate that compared to participants in the individual-AI condition, those in the group-AI condition exhibited a pattern characterised by greater criticality, faster processing and a higher hit rate of high-quality idea adoption when interacting with AI.

4 | Discussion

This study systematically tested how collaboration mode with AI shapes creative performance and whether perspective taking and AI-use strategies mediate that effect. Comparing group-AI, individual-AI and human-only conditions revealed fresh insights into human-AI co-creation and the psychological mechanisms that underpin collaborative innovation. The results inform the design of AI-enhanced collaborative learning environments that foster co-creative student engagement.

Addressing RQ1, we found that group-AI collaboration yielded significantly higher creative performance than both

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individual-AI and human-only groups. Specifically, AI-assisted groups generated ideas with greater usefulness, originality and high-quality fluency. This finding extends prior evidence that AI boosts individual creativity (Doshi and Hauser 2024; McGuire et al. 2024) by demonstrating its amplified impact in group settings. Although individual-AI collaboration raised idea quantity, it lacked the cognitive diversity and dynamic exchange of interpersonal teams, yielding no quality advantage over human-only groups (Medeiros et al. 2025). The pattern supports Hargadon and Bechky's (2006) claim that breakthroughs arise from integrating diverse knowledge during collective engagement, rather than from idea generation alone.

Turning to RQ2, overall human perspective-taking frequency did not differ between group-AI and human-only conditions, but group-AI participants adopted a distinct approach. They seldom copied peer ideas directly and instead reconstructed or elaborated on them with AI as an intermediary. This mediated perspective-taking, with AI reframing peer input, proved a key route to higher creativity. Selective transformation, not imitation, yielded more refined, high-quality outcomes. Two mechanisms may explain this quality gap. First, conversational norms can pressure members to acknowledge low-quality peer ideas, reducing selectivity. Second, AI outputs are typically well-structured and semantically distinct, making them salient creative anchors that are easier to evaluate selectively. These findings extend the cognitive-social-motivational model of group creativity (Nijstad and De Dreu 2012) by showing that AI not only supplies new information but also reshapes the cognitive processes underlying social exchange within teams.

Addressing RQ3, group-AI participants engaged with AI ideas more strategically than individual-AI peers. Both groups adopted AI ideas at similar rates, yet group-AI teams selected high-quality suggestions far more often. Their approach relied on critical, selective evaluation (partial perspective taking) rather than wholesale acceptance. This pattern supports claims that effective human-AI work hinges on strategic engagement, not passive reliance (Hashmi and Bal 2024; Amankwah-Amoah et al. 2024). A likely explanation is distributed cognition in teams, where evaluation, prompting and integration are shared and iteratively refined (Sawyer and DeZutter 2009). Such collective monitoring sharpens critique, improves prompting and maximises AI's creative value. Group-AI teams also produced ideas faster and recorded a higher hit rate of valuable AI contributions, indicating more focused, cognitively aligned interaction. The results align with theories of cognitive alignment and resource coordination in group creativity (Lu and Hao 2026; Lu et al. 2021) and position AI as an external cognitive resource in team processing. Group-AI participants further adopted diverse, stage-specific AI strategies spanning generative exploration, iterative elaboration and final integration. This flexible, stage-sensitive use of AI echoes the Geneplore model (Finke et al. 1992) and highlights AI as a dynamic partner rather than a static tool (Anantrasirichai and Bull 2022).

For RQ4, mediation analyses showed that human perspective taking, AI-directed perspective taking, and strategic AI use jointly linked collaboration mode to creativity. In group-AI teams, less human-to-human perspective taking, which had only a 24% high-quality rate, curbed the spread of weak ideas. An 82% high-quality AI uptake and flexible AI strategies across stages offset this reduction, boosting overall performance. Together the data point to two mechanisms. First, a social-cognitive shift in which peer ideas are elaborated through AI rather than adopted verbatim. Second, cognitive resource optimisation in which humans time AI inputs, gauge idea quality and weave chosen suggestions into the workflow. This dual-path account refines the cognitive-social-motivational model of group creativity (Nijstad and De Dreu 2012; Paulus et al. 2012) and extends evidence that AI reshapes team information processing (Seeber et al. 2020).

At the theoretical level, this study advances the boundaries of group creativity research in several important ways. First, it integrates AI into classical models of group creativity by empirically demonstrating how AI reshapes both interpersonal dynamics and human–AI interaction patterns within teams. Second, by proposing a dual-path model that includes one path involving social-cognitive transformation (e.g., AI-mediated perspective taking) and another involving strategic engagement with AI tools, the findings offer a new conceptualization of how AI reshapes the collaborative process, altering both interpersonal dynamics and individual cognitive strategies within group creativity. This conceptualization is strongly supported by the empirical patterns observed in RQ2 and RQ3, which revealed that AI-mediated perspective taking and flexible strategy use directly contributed to improved idea quality. These findings suggest that creativity in AI-augmented groups emerges not simply from more idea generation, but from the coordinated integration of human and machine contributions. Specifically, this study challenges traditional assumptions that greater perspective taking uniformly benefits creativity, suggesting instead that, in AI-augmented environments, creativity emerges from a combination of selective uptake and strategic engagement with AI. Third, this research provides a paradigmatic case of how group creativity theories can be extended to account for human–AI–human triadic systems, thereby addressing broader questions about how intelligent technologies reconfigure core processes of human cognition, interaction and collaboration.

These findings align with RQ2 and RQ3, where enhanced perspective taking and flexible AI utilisation strategies among learners can be translated into training modules or scaffolded prompts in classroom settings. Instead of positioning AI solely as a resource for individual learners, it may be more effective to integrate AI into group-based learning scenarios, where its generative capabilities can be distributed across learners and tasks. This is, particularly, relevant for open-ended educational activities such as design thinking, problem-based learning or interdisciplinary projects, where idea diversity and exploratory thinking are emphasised. Importantly, the results suggest that the presence of AI does not diminish peer interaction. Rather, when used strategically, AI can support deeper idea processing and more selective integration of both human and machine-generated contributions. For instructional designers and educators, this highlights the potential of aligning AI engagement strategies with different stages of learning tasks. For example, AI can be used during ideation,

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elaboration or refinement phases, depending on the pedagogical goals. These insights also point to the importance of preparing learners not only to work with AI-generated content but to interact with it critically and collaboratively as part of the learning process.

Despite these contributions, several limitations should be noted. First, the study used a structured SDAT, which, while offering clear and controlled measurement, may limit ecological validity. Future research should explore more complex or realistic group tasks (Reich and Teeny 2025). Second, it is worth examining how group-AI collaboration might function in authentic educational settings, such as classrooms, online courses or project-based learning contexts. For example, researchers could investigate how learners across different age groups develop effective strategies for interacting with AI, and whether instructional scaffolds or teacher guidance can facilitate this process. Third, participants in this study were recruited from China, a context characterised by collectivist cultural norms. Prior research suggests that collectivist cultures tend to emphasise continuity with existing solutions over the free exploration of novel ideas (Loewenstein and Mueller 2016). As such, it remains an open question whether the main findings of this study would replicate in more individualistic cultures, where innovation norms and cognitive styles may differ. Future cross-cultural investigations are warranted to examine the generalizability of the current results. Fourth, the sample contained a higher proportion of female participants (approximately two-thirds). Prior studies have not consistently found systematic gender differences in creative performance (Baer and Kaufman 2008; Miller and Halpern 2014). Nevertheless, this gender imbalance may limit the generalisability of our findings, and future research should aim to recruit more gender-balanced samples. Fifth, the experimental setup involved using QQ Chat and ChatGPT 4o side by side in a split-screen arrangement, with the two applications functioning completely independently. While this ensured a clear separation between human communication (via QQ) and AI generation (via ChatGPT 4o), the lack of integration may have introduced minor interface-related constraints (e.g., cognitive switching between windows). Future research could design an integrated platform that combines human dialogue and AI invocation within a single interface, which would more closely resemble real-world collaborative tools and enhance ecological validity. Sixth, although this study focused on material-related innovation, the underlying mechanisms examined, such as perspective taking and AI utilisation strategies, are not domain-specific. However, different disciplines may emphasise different dimensions of creativity. As such, future research is needed to examine whether the current findings generalise across task types and subject areas. Seventh, although AI-based automated scoring provides a standardised and scalable way to evaluate SDAT responses, such scores may still reflect model-specific preferences and are not fully transparent. To support responsible interpretation, we reported human inter-rater reliability and examined human-AI convergence (usefulness $r = 0.86$; originality $r = 0.79$). Future studies could further strengthen validity by using larger panels of expert raters, preregistered scoring rubrics or additional cross-model checks across multiple AI systems and versions. Finally, individual traits, such as cognitive style or openness to technology, should be considered as potential moderators of how people engage in perspective taking and AI strategy use, thereby helping to explain variability in human-AI teaming effectiveness.

5 | Conclusion

This study examined how different modes of human-AI collaboration affect creative performance, with a focus on the mediating roles of perspective taking and AI utilisation strategies. Results showed that group-AI collaboration produced significantly more useful, original and high-quality ideas than individual-AI or human-only collaboration. These effects were mediated by two key psychological mechanisms: (1) AI-mediated transformation of peer ideas, which replaced inefficient direct perspective taking with higher-quality elaboration and (2) strategic deployment of AI across ideation, elaboration and integration stages. Together, these mechanisms enhanced teams' ability to critically evaluate and integrate both human and AI-generated contributions. The findings advance theoretical models of group creativity by positioning AI as a dynamic cognitive partner embedded in distributed human-AI systems. They also offer practical guidance for collaborative learning design by highlighting the value of integrating AI into group contexts where learners can engage with it strategically and collaboratively. Future research should explore how these mechanisms operate in authentic educational settings, across cultures and among learners with different cognitive and technological profiles.

Taken together, this study contributes to the growing knowledge base of collaborative creativity by empirically demonstrating how AI not only enhances individual ideation but also transforms group-level cognitive dynamics through perspective reframing and resource coordination. These findings extend established theories of group creativity (Nijstad and De Dreu 2012) and are supported by recent empirical evidence showing that the integration of generative AI tools significantly enhances team creativity and collaborative problem-solving in educational settings (Wei et al. 2025).

Author Contributions

Zhihao Zhou: conceptualization, investigation, methodology, writing – original draft, data curation, formal analysis, project administration, writing – review and editing. **Letao Xiao:** conceptualization, writing – original draft, validation. **Jiayi Wang:** conceptualization, validation, writing – review and editing, visualisation. **Shuai Tong:** supervision, methodology. **Wenyu Zhang:** conceptualization, writing – review and editing. **Ning Hao:** conceptualization, project administration, supervision, resources, funding acquisition, writing – review and editing.

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Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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